

~~87- $u = f(x)$ $u = a(x+1)$~~

Calculating Second-order partial derivatives.

41- $f(x, y) = x + y + xy$

$$\frac{\partial}{\partial x} x + y + xy$$

$$\frac{\partial}{\partial x} 1 + 0 + y$$

$$\frac{\partial}{\partial x} 0 + 0 + 1$$

$$\frac{\partial}{\partial x^2} 0$$

$$\frac{\partial}{\partial x^2} 0$$

$$\frac{\partial}{\partial y} x + y + xy$$

$$\frac{\partial}{\partial y} 0 + 1 + x$$

$$\frac{\partial}{\partial y} 0 + 0 + 1$$

$$\frac{\partial}{\partial y^2} 0$$

$$\frac{\partial}{\partial y^2} 0$$

$$\frac{\partial}{\partial xy} = \frac{\partial}{\partial y} (1 + y) = 1$$

$$\frac{\partial}{\partial yx} = \frac{\partial}{\partial x} (1 + x) = 1$$

42- $f(x, y) = \sin xy$

$$\frac{\partial}{\partial x} \sin xy$$

$$\frac{\partial}{\partial x} \cos(xy) \cdot y$$

$$\frac{\partial}{\partial x} y \cos(xy)$$

$$\frac{\partial}{\partial x} y \cdot \sin(xy) \cdot y$$

$$\frac{\partial}{\partial x^2} -y^2 \sin(xy)$$

$$\frac{\partial}{\partial y} \sin xy$$

$$\frac{\partial}{\partial y} \cos(xy) \cdot x$$

$$\frac{\partial}{\partial y} x \cos(xy)$$

$$\frac{\partial}{\partial y} x - \sin(xy) \cdot x$$

$$\frac{\partial}{\partial y^2} -x^2 \sin(xy)$$

$$\frac{\partial}{\partial xy} = \frac{\partial}{\partial y} y \cos(xy) = y \cos(xy) - y \sin xy$$

$$\frac{\partial}{\partial xy} = \frac{\partial}{\partial y} y \cos(xy) = (1) \cos(xy) \cdot x \Rightarrow x \cos(xy) - x \sin(xy)$$

$$\frac{\partial}{\partial yx} = \frac{\partial}{\partial x} x \cos(xy) = (1) \cos(xy) \cdot y \Rightarrow y \cos(xy) - y \sin(xy)$$



$$f_{yn} = \frac{\partial}{\partial n} (n \cdot \cos ny)$$

Apply Product rule

$$= uv' + u'v$$

$$= n (-\sin(ny)) \times \frac{\partial}{\partial n} ny + \frac{\partial}{\partial n} n \cdot (\cos ny)$$

$$= -n \sin(ny) \cdot y + 1 \cdot \cos ny$$

$$= -ny \sin(ny) + \cos ny$$

$$f_{ny} = \frac{\partial}{\partial y} (\cancel{n \cos ny}) (y \cos(ny))$$

Apply Product rule

$$= uv' + u'v$$

$$= y (-\sin(ny)) \times \frac{\partial}{\partial y} ny + \frac{\partial}{\partial y} y \cdot \cos(ny)$$

$$= -y \sin(ny) \times n + 1 \cdot \cos(ny)$$

$$= -ny \sin(ny) + \cos(ny)$$

43) $g(m, y) = m^2 y + \cos y + y \sin m$

Sol:-

$$\frac{\partial}{\partial m} m^2 y + \cos y + y \sin m$$

$$\frac{\partial}{\partial m} 2my + (\cancel{\sin 0} + y \cos m)$$

$$\frac{\partial}{\partial m} 2my + y \cos m$$

$$\frac{\partial}{\partial m} 2y + y(-\sin m)$$

$$\frac{\partial}{\partial m} 2y - y \sin m$$

$$\frac{\partial}{\partial y} m^2 y + \cos y + y \sin m$$

$$\frac{\partial}{\partial y} m^2 + (-\sin y) + \sin m$$

$$\frac{\partial}{\partial y} m^2 - \sin y + \sin m$$

$$\frac{\partial}{\partial y} 0 - \cos y + 0$$

$$\frac{\partial}{\partial y} -\cos y$$

Mixed fraction

$$f_{yn} = \frac{\partial}{\partial m} m^2 - \sin y + \sin m$$

$$= \frac{\partial}{\partial m} 2m - 0 + \cos m \Rightarrow 2m + \cos m$$

$$f_{ny} = \frac{\partial}{\partial y} 2my + y \cos m$$

$$\frac{\partial}{\partial y} 2m + \cos m \Rightarrow 2m + \cos m$$

Both are
Same

$$44) h(m, y) = me^y + y + 1$$

Sol:-

$$\frac{\partial}{\partial m} me^y + y + 1$$

$$\frac{\partial}{\partial m} e^y + 0 + 0$$

$$\frac{\partial}{\partial m} e^y$$

$$\frac{\partial}{\partial m} 0$$

$$\frac{\partial}{\partial m^2} 0$$

$$\frac{\partial}{\partial y} me^y + y + 1$$

$$\frac{\partial}{\partial y} me^y + 1$$

$$\frac{\partial}{\partial y} me^y + 1$$

$$\frac{\partial}{\partial y^2}$$

$$\frac{\partial}{\partial y^2} me^y$$

$$f_{yn} = \frac{\partial}{\partial n} me^y + 1 \Rightarrow e^y$$

$$f_{my} = \frac{\partial}{\partial y} e^y \Rightarrow e^y$$

$$45) \alpha(n+y) = \ln(n+y)$$

Sol:-

$$\frac{\partial}{\partial n} = \frac{1}{(n+y)}$$

$$\frac{\partial}{\partial y} = \frac{1}{(n+y)}$$

$$\frac{\partial}{\partial n} = \frac{1}{n+y} \cdot 1$$

$$\frac{\partial}{\partial y} = \frac{1}{(n+y)} \cdot 1$$

$$\frac{\partial}{\partial n} \frac{1}{n+y} \Rightarrow (n+y)^{-1}$$

$$\frac{\partial}{\partial y} \frac{1}{(n+y)} \Rightarrow (n+y)^{-1}$$

$$\frac{\partial}{\partial n^2} = -1(n+y)^{-2} \cdot 1$$

$$\frac{\partial}{\partial y^2} = -1(n+y)^{-2} \cdot (1)$$

$$\frac{\partial}{\partial n^2} = \frac{-1}{(n+y)^2}$$

$$\frac{\partial}{\partial y^2} = \frac{-1}{(n+y)^2}$$

$$f_{yn} = \frac{\partial}{\partial n} (n+y)^{-1} \Rightarrow -1(n+y)^{-2} \Rightarrow \frac{-1}{(n+y)^2} \cdot 1$$

$$f_{ny} = \frac{\partial}{\partial y} (n+y)^{-1} \Rightarrow -1(n+y)^{-2} \times (1) \Rightarrow \frac{-1}{(n+y)^2} \cdot 1$$

$$46- S(n, y) = \tan^{-1}(y/n)$$

Sol:-

$$\frac{\partial}{\partial n} \frac{1}{1 + \left(\frac{y}{n}\right)^2} \times \left(\frac{y}{n}\right)$$

$$\frac{\partial}{\partial n} \frac{1}{1 + \frac{y^2}{n^2}} \times y n^{-1}$$

$$\frac{\partial}{\partial n} \frac{1}{\frac{n^2 + y^2}{n^2}} \times y \cdot 1 y n^{-2}$$

$$\frac{\partial}{\partial n} \frac{\cancel{n^2}}{n^2 + y^2} \times \frac{-y}{\cancel{n^2}} = \frac{-y}{n^2 + y^2}$$

$$\frac{\partial}{\partial n^2} \frac{-y}{n^2 + y^2} \Rightarrow -y (n^2 + y^2)^{-1}$$

$$\frac{\partial}{\partial n^2} -y (-1 (n^2 + y^2)^{-2}) \cdot \partial n$$

$$\frac{\partial}{\partial n^2} y (n^2 + y^2)^{-2} \Rightarrow \frac{\partial n y}{(n^2 + y^2)^2}$$

$$\frac{\partial}{\partial y} \frac{1}{1 + \left(\frac{y^2}{n^2}\right)} \times y n^{-1}$$

$$\frac{\partial}{\partial y} \frac{1}{\frac{n^2 + y^2}{n^2}} \times \frac{1}{n}$$

$$\frac{\partial}{\partial y} \frac{n^2}{n^2 + y^2} \times \frac{1}{n}$$

$$\frac{\partial}{\partial y} \frac{n}{n^2 + y^2} \quad \frac{\partial}{\partial y} n(n^2 + y^2)^{-1}$$

$$\frac{\partial}{\partial y^2} n(-1(n^2 + y^2)^{-2}) \times \partial y$$

$$\frac{\partial}{\partial y^2} \frac{-\partial n y}{(n^2 + y^2)^2}$$

Mixed

$$f_{yn} = \frac{\partial}{\partial n} n(n^2 + y^2)^{-1}$$

Product Rule.

$$= n \cdot -1(n^2 + y^2)^{-2} \cdot \frac{\partial}{\partial n} (n^2 + y^2) + 1 \cdot (n^2 + y^2)^{-1}$$

$$\frac{-n}{(n^2 + y^2)^2} \times 2n + \frac{1}{(n^2 + y^2)}$$

$$\frac{-2n^2}{(n^2 + y^2)^2} + \frac{1}{(n^2 + y^2)}$$